Molecular Potential Energy Surface Mapping via the Variational Quantum Eigensolver (VQE)

A Hybrid Quantum-Classical Framework for Simulating Chemical Bonding & Dissociation Dynamics

ABSTRACT

Accurate characterization of molecular electronic structures and atomic bonding is a foundational pursuit in chemistry, materials science, and drug design. Classical computational quantum chemistry methods face a severe trade-off: mean-field approaches like Hartree-Fock scale polynomially but neglect strong electron correlation, while exact methods such as Full Configuration Interaction (FCI) exhibit exponential Hilbert-space scaling that rapidly becomes intractable. The Variational Quantum Eigensolver (VQE) provides a transformative solution designed specifically for Noisy Intermediate-Scale Quantum (NISQ) devices, combining quantum state preparation with classical numerical optimization.

This project investigates the simulation of molecular potential energy surfaces (PES) across varying interatomic distances to model covalent bond formation, equilibrium geometry, and dissociation phenomena. Using ab initio second-quantized electronic Hamiltonians constructed under the Born-Oppenheimer approximation (via STO-3G and minimal Gaussian basis sets), fermionic operators are transformed into multi-qubit Pauli observables utilizing Jordan-Wigner and Parity mappings. Parameterized quantum circuits—incorporating chemically motivated Unitary Coupled-Cluster Singles and Doubles (UCCSD) and Hardware-Efficient Ansätze (HEA)—are deployed to evaluate expectation values on quantum simulators and hardware, while classical optimizers (e.g., COBYLA, SPSA) iteratively minimize energy observables in accordance with the Rayleigh-Ritz variational principle.

By computing ground-state electronic energies across a continuous grid of nuclear separations for benchmark diatomic and triatomic molecules (such as H_2 and LiH), the computed potential energy curves systematically reproduce critical physicochemical parameters, including equilibrium bond lengths (R_e), harmonic vibrational frequencies, and bond dissociation limits (D_e), achieving near-chemical accuracy compared to exact FCI solutions. Implemented using popular open-source ecosystems such as Qiskit Nature and PennyLane, this investigation demonstrates the viability, visual clarity, and educational power of hybrid quantum-classical algorithms at the intersection of quantum computing and chemical physics.

Keywords: *Variational Quantum Eigensolver (VQE), Potential Energy Surface (PES), Quantum Chemistry, Molecular Bonding, Electronic Structure, Qiskit Nature, PennyLane, NISQ Algorithms.*